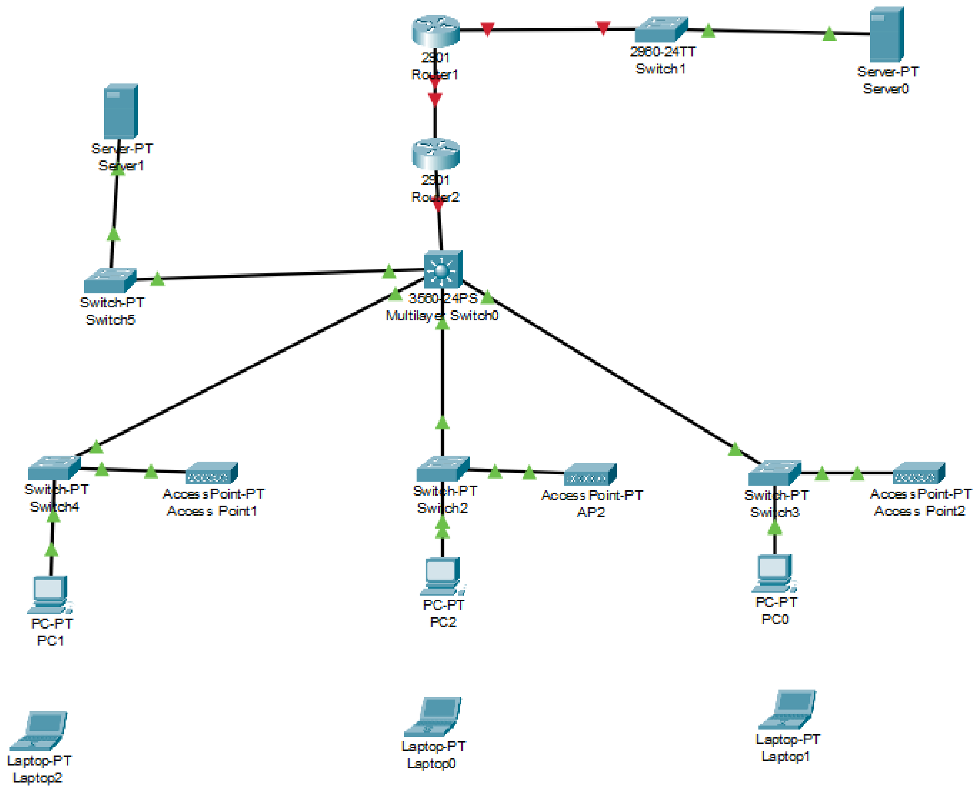
1.Desian Topologi



2. Desian IP Net dan IP Host

No	Nama VLAN	ID VLAN	Subnet IP	Jumlah Host Aktif	Rentang IP Host	Pengguna/Perangkat
1	VLAN-Internet	10	192.168.0.0/24	2	192.168.0.1 – 192.168.0.254	Router1, Server0 (Internet Server)
2	VLAN-Server	20	192.168.1.0/24	2	192.168.1.1 – 192.168.1.254	Router2, Server1 (Lokal Server)
3	VLAN-Core	30	192.168.2.0/24	3	192.168.2.1 – 192.168.2.254	Switch0 (Core), Router1, Router2
4	VLAN-User1	40	192.168.3.0/24	4	192.168.3.1 – 192.168.3.254	PC0, Access Point2, Laptop0, Laptop1
5	VLAN-User2	50	192.168.4.0/24	3	192.168.4.1 – 192.168.4.254	PC1, Access Point1, Laptop2
6	VLAN-User3	60	192.168.5.0/24	3	192.168.5.1 – 192.168.5.254	PC2, AP2, Switch2

3. Desain protokol dn fungsi untuk masing2 perangkat

Perangkat	**Fungsi**	**Protokol yang Digunakan**
Router1 & Router2	Routing antar jaringan/subnet	Static Routing / OSPF / EIGRP
Switch0 (Multilayer)	Core switch, routing antar VLAN jika menggunakan Inter-VLAN	Inter-VLAN Routing
Switch1–Switch5	Distribusi LAN	Ethernet (Layer 2 Switching)
Access Point	Penyedia koneksi wireless ke Laptop	IEEE 802.11, DHCP Relay
Server0	Server internet, DNS, DHCP, Web	HTTP, DNS, DHCP, ICMP
Server1	Server lokal, file sharing, database	FTP, SMB, MySQL, ICMP
PC & Laptop	End device pengguna	DHCP Client / Static IP, HTTP, FTP, DNS

4.Simulasi